

Dormant volcano A volcano that has not erupted for a long time, but could erupt in the future.

Doughnut graphs Graphs that show the relationship of parts to the whole features, e.g. different types of energy contributing to total energy. The numbers can be relative (per cent) or absolute and are added to the graph. Doughnut graphs may have different rings to show different factors or show changes over time, e.g. employment structure.

Drainage basin The area of land drained by a river system.

Drainage density The total length of all the streams and rivers in a drainage basin divided by the total area of the drainage basin.

Drought An extended period of dry weather leading to conditions of extreme dryness. Absolute drought is a period of at least 15 consecutive days with less than 0.2 mm of rainfall. Partial drought is a period of at least 29 consecutive days during which the average daily rainfall does not exceed 0.2 mm.

Economic core region(s) The most highly developed region(s) in a country or in the world.

Economic growth An increase in real GDP (%) after inflation is taken into account.

Economic leakage The part of the money a tourist pays for a foreign holiday that does not benefit the destination country because it goes elsewhere.

Economic periphery Parts of a country (or the world) outside of the economic core region(s).

Economic sustainability Conducting economic activities in a way that preserves and promotes long-term economic wellbeing.

Economy of scale The reduction in unit cost as the scale of an operation increases.

Ecosystem An integrated unit consisting of a community of living organisms (animals and plants) and the physical environment (air, soil, water and climate) that they inhabit. Individual organisms interact with each other and with their habitat.

Ecotourism A specialised form of tourism where people experience relatively untouched natural environments, and ensure that their presence does no further damage to these environments. Ecotourism is at the leading edge of sustainable tourism.

Elderly dependency ratio The ratio of the number of people aged 65 and over to those in the 15–64 age group.

Emigration rate The number of emigrants per thousand population leaving a country of origin in a year.

Energy conservation Reducing the consumption of energy by using it more efficiently.

Energy gap The difference between a country's demand for energy and its ability to satisfy this demand from its own resources.

Energy intensity The amount of energy used to produce a given level of output.

Energy ladder The transition from fuelwood and animal dung to 'higher-level' sources of energy such as electricity.

Energy mix The relative contribution of different energy sources to a country's energy consumption.

Energy pathway A supply route between energy producers and consumers, which may be through pipelines, shipping routes or electricity cables.

Energy policy The actions taken by a government to affect the demand for energy as well as the supply of it.

Energy poverty A lack of access to modern energy services, particularly electricity.

Energy security The uninterrupted availability of energy sources at an affordable price.

Enhanced greenhouse effect (global warming) The additional heat that is trapped as a result of the increased levels of greenhouse gases in the atmosphere. This is largely as a result of human activities and is causing the world climate to become hotter and more extreme.

Environmental impact assessment A study carried out before a development project is undertaken to assess the possible damage to the environment.

Environmental impact survey A document required by law detailing all the impacts on the environment of a construction project above a certain size.

Environmental Performance Index (EPI) An indicator that uses 58 performance indicators across 11 categories to score and rank countries around the globe on climate change performance, environmental health and ecosystem vitality.

Environmental sustainability The preservation and protection of the natural environment over time. It involves appropriate practices and policies to meet present-day needs without compromising the availability of resources in the future.

Epicentre The point on the Earth's surface immediately above an earthquake's focus.

Erosion The wearing away of the Earth's surface by a moving agent, such as a river, glacier or the sea. In a river, there are several processes of erosion, including hydraulic action, abrasion, attrition and solution. In coastal areas, hydraulic action is the most potent form of erosion.

Eutrophication The depletion of oxygen from water in streams and rivers due to increased levels of nitrates (from fertilisers) or phosphates (from detergents).

Evaporation The process in which a liquid turns to a vapour.

Evapotranspiration A combination of the processes of evaporation and transpiration. Vegetation takes in moisture through its root system. It loses some of this into the air by transpiration. Surface water is also lost by evaporation.

Extensive farming Where a relatively small amount of agricultural produce is obtained per hectare of land, so such farms tend to cover large areas of land. Inputs per unit of land are low.

Extinct volcano A volcano that erupted in the past but will not erupt in the future.

Fairtrade A system designed to help small-scale producers in developing countries achieve sustainable and fair trade relationships.

Fertility rate The number of live births per 1000 women aged 15–44 years in a given year.

Fetch The distance of open water over which wind can blow to create waves. The greater the fetch, the more potential power waves have when they hit the coast.

Flash flood A large but temporary increase in channel discharge.

Flood A discharge great enough to cause a body of water to overflow its channel and submerge (flood) the surrounding area.

Flood abatement Reducing the amount of runoff in a drainage basin through actions such as preservation of natural water stores.

Flood diversion The practice of allowing certain areas, such as wetlands and floodplains, to be flooded to a greater

extent.

Flood hydrographs Line graphs that show how the amount of flow in a river (discharge) varies over time following a storm.

Flood impact assessment A process that identifies the impact of a proposed development on the flood risk to the area.

Flood peak The highest discharge of a river flood.

Floodplain The area of periodic flooding along the course of a river valley.

Focus The exact point within or on the Earth's surface where an earthquake occurs.

Food bank A place where stocks of food, typically basic provisions and non-perishable items, are supplied free of charge to people in need.

Food miles The distance food travels from the farm where it is produced to the plate of the final consumer.

Food security When all people at all times have access to sufficient, safe, nutritious food to maintain a healthy and active life.

Food waste Perfectly edible foodstuffs that are thrown away at the retail and consumer level.

Forced migration When people are made to move against their will due to human or environmental factors.

Foreign direct investment (FDI) Investment in physical capital in other countries by transnational corporations.

Fossil fuel A non-renewable resource, such as coal, oil or natural gas, formed from the remains of prehistoric carbon compound- or hydrocarbon-containing materials.

Fraction A fraction is a part of a whole, e.g. $\frac{1}{3}$ of Borneo's rainforest has been deforested.

Four-figure square references A four-figured number that shows the location of a large feature. The first two numbers refer to the eastings and the second two numbers refer to the northings

Freeport A special economic zone in which imported goods can be held or processed free of customs duties before re-export.

Friction The resistance encountered when one body moves relative to another body with which it is in contact.

Fuel poverty When a person on a low-income lives in a home that cannot be kept warm at a reasonable cost.

Fuelwood Wood and charcoal used to supply energy.

Gabion A wire basket filled with rocks or stones, used for stabilising slopes and protecting the base of cliffs in areas of coastal erosion.

Gender dividend The increase in economic growth that can result from greater investment in women and girls, including promoting secondary and tertiary education and female workforce participation.

Geopolitics Political relations among nations, particularly relating to disputes about borders, territories and resources.

Geothermal energy The natural heat found in the Earth's crust in the form of steam, hot water and hot rock.

Glacial period The cold phase in an ice age, i.e. when ice sheets and glaciers advance.

Global city A city that is an important nodal point in the global economic system. Global cities are major financial and decision-making centres.

Global civil society Organisations or individuals, independent from the state, whose aim is to transform policies through communal efforts at a national or global scale.

Global value chain (GVC) A phenomenon whereby production is subdivided into activities carried out in different countries.

Globalisation The increasing interdependence of the world economically, culturally and politically.

GNI per capita The total GNI of a country divided by the total population.

Gorge A narrow, steep-sided valley that may or may not have a river at the bottom.

Green belt An area of largely undeveloped land around a city or town upon which there are wide-ranging planning restrictions on building development.

Green Revolution The introduction of high-yielding seeds and modern agricultural techniques in developing countries.

Greenhouse effect Some atmospheric gases, such as water vapour, carbon dioxide, methane and nitrous oxide, are greenhouse gases. This means that they allow short-wave radiation to pass through the atmosphere and reach the Earth's surface, but they absorb some of the outgoing long-wave radiation thereby heating the Earth's atmosphere. This is a natural process.

Groundwater Water stored underground in a permeable rock, e.g. chalk or sandstone.

Groundwater flow The flow of water through permeable rock.

Growth collapse A significant decline in economic activity brought about by factors such as a global economic shock or by the impact of a major natural hazard.

Groyne A wooden or concrete barrier built at right angles to a beach in order to block the movement of material along the beach by longshore drift.

Guest worker A foreigner who is permitted to work in a country on a temporary basis, for example a farm labourer.

Hard engineering Any coastal or river protection scheme that involves altering the natural environment with concrete, stone, steel, metal, etc. – for example the use of sea walls, gabions, groynes and revetments. Artificial structures are built in order to protect the natural environment from erosion.

Headland A point of land projecting into the sea; also known as a cape or a promontory.

Holocene The most recent geological epoch, beginning 11,700 years ago.

Host culture The culture of a country where a person is currently living or has lived.

Human Development Index The composite UN measure of the disparities between countries, using indicators of health, education and income.

Humidity The quantity of water vapour in a given volume of air (absolute humidity), or the ratio of the amount of water vapour in the atmosphere to the maximum amount the air can hold (relative humidity). At dew point the relative humidity is 100 per cent and the air is said to be saturated. Condensation (the conversion of vapour to liquid) may then occur.

Hydraulic action The erosive force exerted by water alone, such as the sheer force of river water removing loose material from the bed and banks of the river. It is particularly effective on jointed rocks, especially during storm conditions.

Hydraulic radius A measure of the efficiency of a stream's shape – that is, the cross-sectional area divided by the wetted perimeter. The higher the ratio, the more efficient the stream and the smaller the frictional loss. The ideal form is semi-

circular.

Hydrological cycle The movement of water between air, land and sea.

Hydroponics A system where plants may be suspended in nutrient water all the time or fed by an intermittent flow of water.

Immigration rate The number of immigrants per thousand population entering a receiving country in a year.

Impermeable Rocks that do not allow water to pass through.

Import dependency The proportion of energy consumption supplied by imports from other countries.

Industrial estate An area zoned and planned for the purpose of industrial development. Manufacturing industry has a significant presence.

Industrial Revolution The transformation in the late eighteenth and nineteenth centuries of first the UK and then other European countries and the USA from agricultural into industrial nations.

Infant mortality rate The number of deaths of children under one year of age per 1000 live births per year.

Infiltration The initial movement of water from the surface into the upper level of the soil.

Input An element that is required for a process to take place.

Institutional quality The effectiveness, transparency and strength of the institutions within a country, such as the police and the courts.

Intensive farming Agriculture characterised by high inputs per unit of land to achieve high yields per hectare.

Interception The precipitation that is collected and stored by vegetation.

Interglacial period The warm phase during an ice age, i.e. when ice sheets and glaciers retreat.

Interlocking spurs Projecting areas of high ground in the upper course of a river that interlink.

Intermediate technology Aid supplied by a donor country whereby the level of technology and the skills required to service it are properly suited to the conditions in the receiving country.

Intermodal transport hub A facility able to move containers between different modes of transport, with storage space for containers waiting for the next leg of the journey.

Internally displaced person A person who is forced to flee their home due to human or environmental factors, but who remains in the same country.

International aid The giving of resources (money, food, goods, technology, etc.) by one country or organisation to another, lower-income country. The objective is to improve the economy and quality of life in that country.

International tourist arrivals The number of international tourists travelling worldwide or to particular destination countries, usually measured over the course of a year.

International tourist receipts Defined by the WTO as expenditure by international inbound visitors, including their payments to national carriers for international transport.

International trade The exchange of goods and services between countries.

Internet penetration rate The percentage of the total population that use the internet.

Intra-firm trade The international trade of intermediate goods and services within the same firm (company).

Irrigation Supplying dry land with water by systems of ditches or by more advanced means.

Jet stream A narrow band of strong winds that flow from west to east in the Earth's atmosphere.

Kyoto Protocol (1997) An international agreement that aimed to reduce carbon dioxide emissions and the presence of atmospheric greenhouse gases.

Labour intensive A production process where labour costs are high compared to expenditure on capital (factory, machinery, etc.).

Labour migration A migration from one country to another with the primary purpose of seeking employment.

Ladder of agricultural technology Distinct levels of investment in and use of agricultural technology, resulting in an increase in agricultural productivity.

Ladder of development The idea that countries sit on different rungs of a ladder, each associated with a different set of economic activities. As countries develop, they become more capable of producing higher-level goods and services, and move up the ladder.

Lagoon A coastal body of shallow saltwater, usually with limited access to the sea. The term is normally used to describe a shallow sea area cut off by a coral reef or a bar.

Lahar A mudflow consisting of volcanic ash, rain and/or snow.

Landlocked country A country that does not have direct access either to an ocean, or to a sea that is not landlocked.

Land tenure The ways in which land is or can be owned. In the past inheritance laws have had a huge impact on the average size of farms.

Land-use zoning The segregation of land use into different areas for each type of use.

Lateral erosion Sideways erosion of a river bank that results in its valley getting wider.

Latitude Lines parallel to the equator that indicate how far north or south of the equator a place is, e.g. Cairo is located at approximately 30°N.

Lava flow A slow-moving flow of molten lava, typically moving at 1–10 km/hour and with temperatures of over 700°C.

Least low-income countries The very poorest countries.

Levéé A raised banks found along the side of a river channel.

Life expectancy at birth The average number of years a newborn infant can expect to live under current mortality levels.

Line of best fit A line that summaries the main features of a scatter graph. It does not have to pass through the origin or all points, but should show the main relationship shown by the graph.

Load The particles of sediment and dissolved matter carried along by a river.

Long profile A longitudinal section of the course of a river drawn along the river from source to mouth.

Longitude Lines running from the north pole to the south pole that indicate how far east or west of the Greenwich Meridian (London) a place is, e.g. Cairo is located at approximately 31°E.

Longshore drift The movement of material along a beach by wave action. When a wave breaks obliquely (at an angle to the beach), pebbles are carried up the beach in the direction of the wave (swash). The wave returns to the sea (backwash) at right angles to the beach (direction of steepest slope), carrying material with it.

Magma Molten or semi-molten material found beneath the Earth's surface.

Magma chamber An underground reservoir of molten magma.

Malnutrition Insufficiency in one or more of the nutritional elements necessary for health and wellbeing.

Managed retreat When the coastline is allowed to retreat (erode) in certain areas where the population density or the value of land is low, so that nature takes its course.

Mangrove A salt-tolerant forest of trees and shrubs that grows in the tidal estuaries and coastal zones of tropical areas.

Marginalisation The process of being pushed to the edge of economic activity.

Marine protected area (MPA) An area of coastline or ocean that is given some form of protection or conservation, e.g. activities such as fishing, mining or drilling may be banned or restricted.

Mass migration The migration of a large group of people from one geographical area to another.

Median The middle value when all the data are placed in ascending or descending order.

Mean A type of average; it is the sum of all the values divided by the number of values.

Meander neck A thin piece of land between two successive outer bank meanders, likely to be eroded in the future to form an oxbow lake.

Mechanisation The replacement of human or animal forms of power by a mechanical system such as a tractor in agriculture.

Mercalli Scale A scale that measures the effects of an earthquake based on its intensity of shaking and the observed damage.

Microcredit Tiny loans and financial services to help people, mostly women, start businesses and escape poverty.

Microgeneration A generator that produces electricity with an output of less than 50 KW.

Migrant culture The attitudes and values of a particular society to the process of migration.

Migration The movement of people across a specified boundary, national or international, to establish a new permanent place of residence.

Mitigation (of climate change) Measures used to prevent or limit global climate change, e.g. burning less fossil fuel and/or planting more trees.

Mixed farming Cultivating crops and keeping livestock together on a farm.

Mode The most common value in a population.

Moment Magnitude Scale A scale that measures the amount of energy released by an earthquake.

Monsoon The season of heavy rain during the summer in hot Asian countries.

Mouth The point at which a river flows into a much larger body of water – an ocean, sea or lake.

Multiplier effect The idea that an initial amount of spending or investment causes money to circulate in an economy, bringing a series of economic benefits over time.

Natural hazard A natural event that puts people, property and livelihoods at risk.

Negative indices Powers that have a minus sign, e.g. $2^3 = \left(\frac{1}{2}\right)^{-3} = \frac{1}{8}$

Negative relationship A negative relationship occurs when one value decreases while the other increases.

Neo-colonialism The dominance of lower-income countries by wealthy ones, not by direct political control (as in colonialism), but by economic power and cultural influence.

Net migration The difference between the numbers of immigrants and emigrants in a year.

New Green Revolution An approach to increasing agricultural production that draws upon the best of the agricultural technologies currently available. It emphasises improved farm management in order to minimise environmental damage from external inputs and benefits farmers and marginal areas bypassed by the original Green Revolution.

New international division of labour (NIDL) Divides production into different skills and tasks that are spread across countries rather than taking place within a single country.

Newly industrialised country A country that since the 1960s has experienced very rapid growth in manufacturing industry.

Non-renewable energy The fossil fuels (coal, oil, natural gas) and nuclear fuel. These resources are finite so that as they are used up the supply that remains is reduced. Eventually, these non-renewable resources could become completely exhausted.

Nutrient cycling The re-use of nutrients over time. They are recycled between the atmosphere, plants, animals and the soil.

Open economy A country with few investment and trade barriers, which encourages business with other countries.

Organic farming Agriculture that does not use manufactured chemicals such as chemical fertilisers, pesticides, insecticides and herbicides.

Output A finished product that is sold to customers or consumed, for example what a farm produces, such as milk, eggs, meat and crops.

Overgrazing The grazing of natural pastures at stocking intensities above the livestock carrying capacity.

Overland flow Water flowing over the surface under the influence of gravity. It occurs when the soil is saturated.

Oxbow lake A curved lake found on the floodplain of a river. Oxbows are caused by the loops of meanders being cut off at times of flood and the river subsequently adopting a shorter course.

Package tour The most popular form of foreign holiday, where travel, accommodation and meals may all be included in the price and booked in advance.

Paris Agreement A legally binding international treaty on climate change. It was adopted by 196 Parties in 2015 and came into force in 2016. Its aim is to keep the increase in the global average temperature to well below 2°C above pre-industrial levels.

Pastoral farming The rearing of livestock, such as dairy cattle, sheep and pigs.

Per capita calorie supply The main international indicator used to show variations in global access to food supply. It has limitations – a wholesome diet requires more than just energy and supply figures do not include consumption-level waste (food that is wasted at household, restaurant and retail levels). However, it is the best indicator available. It is measured in kilocalories per person per day.

Percentage A number or ratio expressed as a fraction of 100 (%).

Percolation The downward, vertical movement of water within soil or rock.

Permeable Rocks that allow water to pass through, either due to porosity (large volume of pore spaces) or due to being

pervious (having joints and cracks/fissures).

Plantation A large farm or estate where one crop is produced commercially, such as palm oil in Malaysia or tea in Sri Lanka. Plantations are usually owned by large companies, often multinational corporations. Many plantations were established in countries under colonial rule, using slave labour.

Pleistocene The geological era lasting from about 2.6 million years ago to around 11,700 years ago, characterised by multiple glacial (cold) periods and interglacial (warm) periods.

Pollution Contamination of the environment.

Population momentum When a country's fertility rate declines to or below the replacement level, but the population continues to grow because of its age structure.

Population projection A prediction of the future population, usually based on current and past trends.

Population pyramid A bar chart, arranged vertically, that shows the distribution of a population by age and gender.

Population structure The composition of a population, the most important elements of which are age and sex (gender).

Positive index A power value that is positive, e.g. $2^3 = 2 \times 2 \times 2 = 8$.

Positive relationship A positive relationship exists between two values when both increase/decrease together.

Post-industrial society A country where the tertiary sector dominates employment and where considerable deindustrialisation has occurred.

Pothole A circular depression in a river bed formed by the abrasive action of rock particles caught up in vortices in turbulent river flow.

Precipitation Water that falls to the Earth from the atmosphere. It is part of the hydrological cycle. Forms of precipitation include rain, snow, sleet, hail, dew and frost.

Preservation Maintaining a location exactly as it is and not allowing development.

Primary product dependent (commodity dependent) When a country relies on one or a small number of primary products for most of its export earnings.

Primary sector The economic sector that exploits raw materials. Farming, fishing, forestry and mining make up most jobs in this sector.

Primary wave The fastest type of seismic wave, which causes the ground to shake.

Process An operation that takes place on a farm, such as ploughing and harvesting, or the industrial activities that take place in a factory to make a finished product.

Product chain The full sequence of activities needed to turn raw materials into a finished product.

Productive capability The productive resources, entrepreneurial capabilities, and production links that together determine a country's capability to produce goods and services, and enable it to grow and develop.

Pro-natalist policy A policy that is encouraging and supportive of increasing the fertility rate.

Pro-poor tourism Tourism that results in increased net benefits for lower-income people.

Proportional symbols Symbols such as squares, circles and cubes that are drawn relative to the features being looked at, e.g. proportional circles to show the size of world megacities.

Protectionism The practice of shielding a country's domestic industries from foreign competition by taxing imports.

Protozoans A group of single-celled organisms, either free-living or parasitic, that feed on organic matter or organic debris.

Pull factor A positive factor at the destination.

Purchasing power parity (PPP) Takes into account variations in the cost of living between countries.

Push factor A negative factor at the place of origin.

Pyroclastic flow A superheated cloud of volcanic ash, cinder, tephra and other fragments. It can reach temperatures of over 700°C and is capable of travelling at speeds of over 500 km/hour.

Quaternary sector The economic sector that uses high technology to provide information and expertise to all sectors of an economy.

Rainfall A form of precipitation in which drops of water fall to the Earth's surface from clouds. The drops are formed by the accumulation of fine droplets that condense from water vapour in the air. The condensation is usually brought about by the rising and subsequent cooling of air.

Range The difference between the highest value and the lowest value.

Range How far people will travel to obtain a good or service, e.g. a short distance for a newspaper (a low-order good), or a large distance for a car/computer (a high-order good).

Rapid Where very thin alternating bands of hard and soft rock cross the course of a river, creating an uneven river bed and a zone of turbulent water.

Rate of natural change The difference between the birth rate and death rate. If it is positive, it is termed natural increase. If it is negative, it is known as natural decrease.

Ratio A method of comparing relative size or proportions, e.g. the area deforested in Borneo compared to the forested areas is 1:2 (i.e. one-third has been deforested and two-thirds remain).

Refugee A person who is forced to flee their home due to human or environmental factors and who crosses an international border into another country.

Remittance Money sent back by a migrant to their family in their home community.

Renewable energy Sources of energy such as solar and wind power that are not depleted as they are used.

Replacement level fertility A fertility rate of 2.1 children per woman is required to maintain a population at its current level. Below this level, a population will eventually decline.

Repowering Replacing first-generation wind turbines with modern multi-megawatt turbines, which give a much better performance.

Reurbanisation The increase in population in the inner city of an urban area following a period of urban decline.

Richter Scale A scale to measure the magnitude of an earthquake. The scale is logarithmic, so an earthquake of 5.0 magnitude is ten times greater than one of 4.0 magnitude.

River cliff A steep slope that may be found on the outer bend of a meander.

River terrace A relatively flat river deposit marking the position of a former floodplain.

Root zone The area of soil and oxygen surrounding the roots of a plant.

Route maps Maps that show the main transport elements in an area, e.g. roads, rail lines, or maps that show the

fastest/shortest route between two places.

Saffir-Simpson Scale A scale to measure the intensity, windspeed and likely damage caused by tropical cyclones.

Saltation When larger particles of a river's load (sand, gravel, very small stones) are transported in a series of 'hops' or bounces along the river bed. It is the means by which bedload (material that is too heavy to be carried in suspension) is transported downstream.

Sand dune A mound or ridge of wind-drifted sand common on coasts and in deserts. In coastal areas, sand is trapped by vegetation, notably sea couch grass and marram grass, to form stable dunes.

Sanitary landfill Municipal waste facilities designed to control leachate and methane, thus minimising the risk of land pollution from solid waste disposal. Sanitary landfill sites are prepared with impermeable bottom liners to collect leachate and prevent contamination of groundwater.

Saturated When the soil is so full of water that it cannot absorb any more.

Scales Scales refer to the relationship between the distance on a map and the corresponding real-life distance on the ground, e.g. on a 1:25,000 map, every 4 cm on the map refers to 1 km on the ground, and on a 1:50,000 map every 2 cm on the map refers to 1 km on the ground.

Sea floor spreading The gradual increase in the size of the oceans as new material is added to the oceans at constructive plate boundaries/mid-ocean ridges.

Sea level An average level of the sea, between the high water mark and low water mark.

Seasonality Fluctuation in demand in the tourism industry over the course of a year, due mainly to seasonal variations in the weather.

Secondary cone A smaller cone found on the side of a volcano's main cone.

Secondary sector The economic sector that manufactures primary materials into finished products.

Secondary wave A secondary or shear wave is a seismic wave that travels through solids, with an undulating/wavelike movement (up and down).

Seismic wave A wave-like motion through the Earth caused by an earthquake, volcanic eruption or landslide.

Sequester To capture and remove CO₂ from the Earth's atmosphere and to store it permanently.

Shallow-focus earthquake An earthquake that occurs up to a depth of 70 km.

Shelterbelt A barrier of trees and shrubs that protects against the wind and reduces erosion.

Shield volcano A low-angle volcano formed of very hot, runny, basaltic lava.

Shifting cultivation A farming system in which farmers move on from one place to another when the land becomes exhausted. The most common form is slash-and-burn agriculture, where land is cleared by burning so that crops can be grown. After a few years the soil fertility is reduced and the land is abandoned. A new area is cleared while the old land recovers its fertility.

Six-figure references A set of six numbers that identifies the exact location of a place on a map. The first three numbers refer to the eastings and the second three numbers refer to the northings.

Slip-off slope The gentle slope found on the inside bank of a river meander.

Slow travel Visitors seeking a deeper connection to the culture, history and nature of the place they are visiting. They usually stay for longer at a destination while avoiding crowds of tourists and appreciating lesser-visited attractions.

Smart meter An internet-capable device that records the detailed use of household energy.

Smart power grid An electricity network that uses digital and other advanced technologies to monitor and manage the transport of electricity from all generating sources to meet the varying electricity demands of end users.

Social business A form of business that seeks to profit from investments that generate social improvements and serve a broader human development purpose.

Social norm The general attitude of a population to an important issue, such as family size, contraception, religion, politics, etc.

Social sustainability The wellbeing of people and the communities they live in.

Soft engineering Any form of coastal or river protection that involves the use of natural means, e.g. sand dunes, saltmarshes, tree planting and/or beach replenishment.

Soil The outermost layer of the Earth's solid surface, consisting of weathered rock, air, water and decaying organic matter overlying the bedrock. Soil comprises minerals, organic matter (humus) derived from decomposed plants and organisms, living organisms, air and water.

Soil erosion The wearing away and redistribution of the Earth's soil. It is caused by the action of water, wind and ice, and also by unsustainable methods of agriculture.

Solution (or corrosion) The process by which the minerals in a rock, notably calcium ions, are dissolved in acid water. Solution is one of the processes of erosion.

Source The origin or starting point of a river.

Sphere of influence maps Maps that show the influence of a particular function, i.e. how far people will travel to a place/service such as a settlement or a shop. Some spheres of influence are very small, e.g. a small hamlet, whereas others may be international, e.g. the sphere of influence of a world city.

Spit A ridge of sand or shingle connected to the land at one end and the open sea at the other end. It is formed by the interruption of longshore drift due to wave refraction, river currents, secondary winds and/or changes in the shape of the coastline.

Stack An isolated, upstanding pillar of rock that has become separated from a headland by coastal erosion. It is usually formed by the collapse of an arch.

Standard notation The number that we would normally write, e.g. 281.

Stemflow Water that trickles down plant stems and tree trunks.

Steric effect The expansion of sea water as its temperature increases.

Store (of water) A body of water that receives, holds and releases volumes of water. On land, these include rivers, lakes, reservoirs and aquifers.

Strategic petroleum reserve A large reserve of oil held by a country to tide it over for a few months or so if normal oil supplies are disrupted.

Stratovolcano A cone-shaped volcano formed of viscous, sticky lava.

Structural transformation The transition of an economy from low productivity and labour-intensive activities to higher

productivity and skill-intensive activities.

Stump An eroded stack that is exposed only at low tide.

Subduction zone An area where a dense oceanic plate plunges underneath a less-dense continental plate.

Subsistence agriculture The most labour-intensive and generally the smallest scale of agriculture, where the produce is consumed entirely or mainly by the family who work the land or tend the livestock.

Subsistence farming The most basic form of agriculture, in which the produce is consumed entirely or mainly by the family who work the land or tend the livestock. If a small surplus is produced, it may be sold or traded.

Suburbanisation The growth of residential areas at the edges of urban areas; the movement of people into the suburbs (outer parts of the city).

Suburb A residential area found towards the edge of a city and located relatively far from the central areas.

Supervolcano A volcano with a Volcanic Explosivity Index of 8.0, i.e. an eruption of over 1000 km³.

Surface runoff The unconfined flow of water over the ground surface. It occurs when excess precipitation can no longer infiltrate into the soil.

Surface wave A seismic wave at the Earth's surface. Love waves cause the ground to move sideways, and Rayleigh waves cause the ground to move up and down.

Suspension When the smallest and lightest particles of a river's load (silt and clay) are carried in suspension by the moving water.

Sustainable development Development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

Sustainable Development Goals (SDGs) The UN's 17 Sustainable Development Goals with their 169 targets were adopted by UN Member States in 2015. The overall objectives were to end poverty, protect the planet, and ensure all people enjoy peace and prosperity.

Sustainable drainage system (SuDS) A natural drainage method for the drainage of surface water to nearby watercourses in urban areas in particular.

Sustainable tourism Tourism organised in such a way that its level can be sustained in the future without creating irreparable environmental, social and economic damage to the receiving area.

Swash The movement of material up the beach in the direction of the prevailing wind.

System A situation in which there are recognisable inputs, processes and outputs, such as raw materials, labour, energy and capital. For example, a farm requires a range of inputs, such as labour and energy, before anything else can happen.

Technology transfer The transfer of new technology, especially from developed to developing countries, in an attempt to boost their economies.

Tephra All volcanic fragments ejected into the air during an eruption.

Terracing Created when a steep slope is converted into a series of flat steps with raised outer edges (bunds). The monsoon regions of Southeast Asia exhibit widespread terracing.

Tertiary sector The economic sector that provides services to people and to businesses.

Thermal limit for crop growth The temperature (low or high) beyond which a plant is unable to grow to a healthy state.

Threshold characteristic The number of people needed to support a service, e.g. a small number for a newsagent, a large number for a car salesroom.

Throughflow The flow of water through the soil under gravity.

Tied aid Foreign aid that must be spent in the country providing the aid (the donor country).

Tiger economy An economy that undergoes rapid economic growth, usually accompanied by an increase in the standard of living.

Tipping point A critical threshold when even a small change can have dramatic effects and cause a disproportionately large response.

Tombolo A bar or ridge that links an island to the mainland.

Total fertility rate The average number of children born to a woman in a country during her lifetime.

Tourism Travel away from the home environment for leisure, recreation and holidays, or to visit friends and relatives, or for business and professional reasons.

Tourist-generating country A country from which many people take holidays abroad.

Traction The sliding or rolling of sediment along the river bed or sea floor.

Trade bloc A group of countries that share trade agreements between each other.

Transfer (of water) The movement of water between stores in the hydrological cycle.

Transnational corporation (TNC) A firm that owns or controls productive operations in more than one country through foreign direct investment.

Transpiration The loss of moisture from vegetation into the atmosphere.

Transport hub A location where routes converge, allowing for the efficient movement of goods and people. The most important transport hubs are often intermodal.

Transportation The movement of a river's load by the processes of traction, saltation, suspension and solution.

Travel motivator The reasons that people travel. The three major categories are leisure, business and visiting friends and relatives.

Triangular graph An equilateral triangle with three axes. Points can be plotted by reading the correct value off each of the axes and placing a point on the triangular grid.

Tributary A river or stream flowing into a larger river or lake.

Tropical cyclone (hurricane, typhoon) A region of very low atmospheric pressure in tropical regions. Hurricanes originate in latitudes between 5° and 20° north or south of the equator, when the surface temperature of the ocean is above 27°C. A central calm area, called the eye, is surrounded by inwardly spiralling winds (anticlockwise in the Northern Hemisphere) of up to 320 km/hr.

Tropical rainforest Dense forest usually found on or near the equator, where the climate is hot and wet. The vegetation in tropical rainforests typically includes a canopy formed by high branches of tall trees that provides shade for lower layers, an intermediate layer of shorter trees, tree roots and lianas, and a ground cover of mosses and ferns.

Tsunami A large and unusual wave in the water generated by a submarine earthquake, volcanic eruption or landslide.

Ultra Low Emission Zone (ULEZ) A scheme in London that has significantly reduced the number of older, high-polluting

vehicles entering the city's urban area.

Ultra-large container ship A vessel with a capacity of at least 14,000 TEUs (twenty-foot equivalent units).

Undercut Erosion taking place at the base of a river bank or waterfall and leading to the formation of a notch and an overhang.

Unicorn A recently established business (start-up) valued at US\$1 billion or over.

Unimodal A distribution of a population where there is one clear most common size.

United Nations Commission for Trade and Development (UNCTAD) The UN's commission that deals with trade, investment and development issues, assisting countries to integrate into the world economy on an equitable basis. It is based in Geneva, Switzerland.

United Nations Development Programme (UNDP) The UN's development agency working to eradicate poverty. It plays a crucial role in helping countries achieve the Sustainable Development Goals.

United Nations Framework Convention on Climate Change (UNFCCC) The UN process for reaching an agreement to limit dangerous climate change. Its main objective is the stabilisation of greenhouse emissions to prevent dangerous human-caused climate change.

Urban growth The absolute growth of urban areas in population size and/or the physical growth of urban areas.

Urbanisation An increase in the proportion of people living in urban areas.

US Geological Survey A United States scientific agency that studies the landscape of the USA, its natural resources, and the natural hazards that threaten it.

Velocity The speed of a river's water, measured in metres per second.

Venn diagrams Diagrams that show the relation (or overlap) between different sets of data, e.g. reasons for rate of development in LICs.

Vent A fracture on the summit of a volcano through which lava can be ejected.

Vertical erosion Downward erosion occurring in the upper course of the river, where the river cuts down into its bed, deepening the valley.

Volcanic block A fragment of rock measuring over 64 mm in diameter that is erupted in a solid condition.

Volcanic Explosivity Index A measure of the amount of volcanic material ejected in an eruption, the height the material is thrown into the atmosphere, and how long the eruption lasts. The scale is logarithmic, so a VEI 6.0 is ten times greater than a VEI 5.0.

Volcanic gas A gas given off by an active volcano, including water vapour, carbon dioxide, sulphur dioxide, hydrogen sulphide, nitrogen, methane and carbon monoxide.

Volume The amount of water in the river.

Voluntary migration When individuals have a free choice about whether to migrate or not.

V-shaped valley Rivers in upland areas contain large boulders that can erode the bed rapidly when the river is in flood. This results in the river cutting downward into its bed by vertical erosion to form steep V-shaped valleys.

Water quality The suitability of water to sustain various uses or processes. Any use will have certain requirements for the physical, chemical or biological characteristics of the water.

Water table The upper level of groundwater saturation in permeable rocks.

Water treatment Improving water quality to make it acceptable for a particular end use, such as human consumption or irrigation.

Waterfall A steep fall of river water where its course is suddenly and significantly interrupted.

Watershed A ridge of high land that forms the boundary between two drainage basins.

Wave The circular or elliptical movement of water near the surface of the sea.

Wave refraction The way in which a wave changes shape and loses speed as it comes into contact with the sea bed. If refraction is complete, waves break parallel to the coastline. If refraction is not complete, longshore drift occurs.

Wave-cut platform A gently sloping rock surface found at the base of a coastal cliff. It is covered by water at high tide but is exposed at low tide. It is formed by the erosion (by waves) of a former cliff face.

Weight gain Where the finished product is heavier than the weight of the raw materials required to manufacture it.

Weight loss Where the finished product is lighter than the weight of the raw materials required to manufacture it.

Wellfield The land above and surrounding wells drilled into an aquifer.

Wetted perimeter The total length of the cross-section at the interface between a channel bed and the stream water that occupies it.

Wind roses Diagrams that show the direction and wind flow for a certain time period. More complex ones can also show variations in wind over the same time.

Youth dependency ratio The ratio of the number of people aged 0–14 years to those in the 15–64 age group.

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Ickes, 'The Mississippi River: A Place for Fish' p.6, www.researchgate.net/publication/267897610; **p.32** Figure 1.47 Baker, S. et al. 1996, *Pathways in Senior Geography*, Nelson, Figure 5.3, p.88; **p.33** Figure 1.48 www.researchgate.net/figure/Water-flow-from-Blue-Nile-and-Nile-River-before-and-after-the-GERDTesfa-2013_fig7_336591659; **p.37** Figure 2.5 G. Nagle, AS & A2 *Geography for Edexcel B* (Oxford University Press, 2003) copyright © Garrett Nagle 2003; **p.40** Figure 2.14 (t) Republic of South Africa/Department of Land Affairs/Surveys and Mapping; **p.41** Figure 2.15 G. Nagle, *ORG GCSE Geography* (Through Diagrams) (Oxford University Press, 1998), copyright © Garrett Nagle 1998; **p.59** Figure 2.35 adapted from S. Warn and C. 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Harcourt and S. 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Guinness, *Geography for the IB Diploma: Patterns and Change* (Cambridge University Press, 2010), © Cambridge University Press 2010; Figure 8.9 <https://datatopics.worldbank.org/worlddevelopment-indicators/the-world-by-income-and-region.html>; **p.185** Figure 8.10 <https://datatopics.worldbank.org/world-development-indicators/stories/the-classification-of-countries-by-income.html>; Table 8.6 World Bank; **p.188** Figure 8.13 P. Guinness, *Geography for the IB Diploma: Patterns and Change* (Cambridge University Press, 2010), © Cambridge University Press 2010; **p.191** Figure 8.15 P. Guinness and B. 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Index

A

- abrasion 10, 35
- Accumulated Cyclone Energy (ACE) 54
- Achill Island 43
- active volcanoes 95
- adaptation 118–19
- adult literacy rate 180
- advection 7
- age, population structures 134, 136, 137–9
- agglomeration 216–17
- agriculture
 - and climate change 114
 - and coastal areas 48
 - subsistence 210
 - sustainable 194
 - and urbanisation 166
 - see also* farming
- agro-industrialisation 253
- aid 198–200, 263–4
- air freight 223
- albedo 68
- alluvium 11
- Amazon River 2–3
- Angel Falls 14
- annotation 292–3, 297, 317
- Antarctic Circle 67
- Antarctic Treaty 1959 74
- Antarctica 7, 67
 - abiotic factors 70–1
 - biotic factors 70–1
 - climate 67–8
 - climate change 72, 73
 - ecosystem 67, 69–75
 - fishing 72–3
 - ice sheets 67, 72, 73
 - management strategies 74–5
 - pollution 73
 - threats to 72–3
 - tourism 73
- aquaculture 48
- aquaponics 249
- aquifers 8–9
- arable farming 247, 253–4
- area, estimating 293
- artificial intelligence (AI) 225
- ash fallout 96
- asylum seekers 142, 152
- atlas maps 291–2
- atoll reefs 59
- attrition 10, 36
- Australia, offshoring asylum seekers 152
- averages 307–8

B

- backwash 36
- Bangladesh, impacts of climate change 120–1
- bar graphs 300–1
- barrier reefs 59
- base maps 292–3
- battery energy storage systems (BESS) 286
- bays 39
- beaches 41–3, 45
 - see also* coastal areas
- bedload 11, 36
- bilharzia 29
- biodiversity 77
- biofuels industry 80

- biomass 283–4
- birth rate 127, 128–33
- Bradshaw model 5
- braided channels 16
- Brazil, deforestation 81–3
- break-of-bulk points 214
- Butler Model 235–6
- butterflies 79

C

- calcium carbonate 36
- calorie intake 181
- carbon dioxide 111, 114
- Caribbean Sea 53
- cartographic skills 291–8
- cartoons 310
- cattle ranching 80
- caves 39–40
- chlorofluorocarbons 114
- cholera 29
- choropleth maps 297
- cinder cone volcanoes 95
- circular economy 193–4
- Clark-Fisher economic sector model 210
- clean air transition 271
- cliff drainage 50
- cliff regrading 51
- cliffs
 - coastal areas 39, 41
 - rivers 14
- climate
 - Antarctica 67–8
 - impact on rivers 11
 - tropical rainforests 76–7
- climate change
 - adaptation and mitigation 118–19
 - in Antarctica 72–3
 - evidence for 110–12
 - and flooding 20–1
 - global warming 72, 110, 115
 - greenhouse effect 113–14, 115–16
 - human influences 113–14
 - impacts of 115–17
 - management strategies and responses 118–21
 - natural causes 112–13
 - orbital changes 112–13
 - tipping point 115
 - and volcanic activity 113
- climate loss 261
- coal 214, 274, 279
- coastal areas
 - bars and spits 43–4
 - beaches 41–3, 45
 - concordant/discordant coastlines 46
 - erosion 35–8, 49, 57
 - fieldwork 63–4
 - hazards 48–9, 54–5
 - landforms 35, 39
 - living near 47–9
 - management 49–51
 - pollution 49
 - sand dunes 45
 - stacks and stumps 39–40
 - waves 35, 37–40
- colonialism 188
- commercial farming 247–8
- commodity chains 221–2
- communications systems 224
- condensation 8
- cone volcanoes 95
- conservation 74
- construction industry 195–6

- constructive waves 37
- consumer culture 224
- convection currents 88–9
- convectonal rains 76–7
- coral reefs 58–60, 62–3
- Coral Triangle 62–3
- corrasion 10, 35
- corrosion 13, 36
- cotton crops 259
- counterurbanisation 158
- crater 96
- cultural diffusion 223–4
- cultural identity 151
- culture
 - and development 190
 - and globalisation 223–4

D

- Danube River 3
- Danum Valley Conservation Area 84
- data collection 312
- death rate 127, 128, 132–3
- debt relief 200
- decentralisation 167
- decimals 308
- deforestation 20, 81–2
- deindustrialisation 211
- deltas 3, 16–17
- demographic change 125
- demographic dividend 137
- dependency theory 188–9
- deposition 6, 11, 15, 16, 36
- desertification 266–7
- deserts 9
- destructive waves 37
- development 178
 - categories of 179
 - and culture 190
 - dependency theory 188–9
 - economic indicators 179–82
 - gap 187–8
 - high-income countries (HICs) 184–5, 187
 - Human Development Index (HDI) 182–3
 - ladder of 184
 - low-income countries (LICs) 184–6
 - middle-income countries MICs) 184–5, 186–7
 - sustainable 191–6
 - and technology 190
 - trade and investment 189–90
 - uneven 184–90, 196–205
- diaspora 149
- digital divide 190
- digital twins 225
- dispersion graphs 303
- disruptive technology 225
- distribution maps 298
- doctors, per thousand people 181–2
- dormant volcanoes 95
- doughnut graphs 305
- drainage basins 2–3, 7

E

- Earth
 - crust 87
 - inner core 87
 - layers and characteristics 87
 - lithosphere 87
 - mantle 87
 - outer core 87
 - plate boundaries 89–94
 - tectonic plates 88
- earthquakes 91–4, 98–106

- economic development 179–82
- economic migration 142, 144, 146
- economic sectors 208–10, 212
- economic sustainability 194
- Ecosystem Approach to Fisheries Management (EAFM) 63
- ecotourism 241
- education 165, 180
- electric vehicles 195
- emigration rate 127
- employment 163
- employment structures 210
- energy
 - conservation 286
 - consumption 272–5
 - gap 276
 - mix 271, 274
 - non-renewable 270–1, 279–81
 - poverty 271
 - production 272–5, 277–8, 279
 - renewable 270–1, 274, 281–5
 - security 276
 - storage 286
 - supply 276–7, 285–6
 - surplus and deficit 276–7
- enhanced greenhouse effect 114, 115–16
- environment
 - and tourism 239–41
 - and urbanisation 170
- environmental impact surveys 74
- Environmental Performance Index (EPI) 192–3
- environmental sustainability 193
- equatorial areas 76–7
- erosion
 - coastal 35–8
 - rivers 5, 10–11, 14–15
- European Union (EU) 222
- eutrophication 29
- evaporation 7–8
- evapotranspiration 9
- extinct volcanoes 96

F

- Fairtrade 198
- farming
 - aquaponics 249
 - arable 247, 253–4
 - commercial 247–8
 - extensive 249
 - hydroponics 249
 - intensive 249
 - mixed 247
 - no-till 266
 - organic 249
 - pastoral 247, 253–4
 - subsistence 247–8
 - sustainable 264–5
 - systems 250
 - see *also* agriculture
- fault lines 13
- Favela Barrio Project 164–5
- fertility 128–31
- field sketches 316–17
- fieldwork
 - coastal areas 63–4
 - skills 310–20
- fisheries 48
- fishing 62, 72–3
- flash floods 20, 57
- flood hydrographs 305
- flooding 115, 121
 - causes of 19–20, 26
 - and erosion 11

- flood peak 82
- impacts of 18–19, 20–1, 26, 57–8
- lake formation 15
- Mississippi River 25–7
- precipitation 26–7
- prediction and prevention 22–4
- risk management 21
- floodplains 15–16, 20
- flow-line diagrams 302–3
- food aid 263–4
- food production 116–17, 252–5
- food security 261–4, 268–9
- food supply
 - challenges 258–64, 268–9
 - global variations 251–3
 - globalisation of 255–7
 - strategies 254–5
- food waste 252
- Foote, Eunice Newton 111
- fractions 308
- Frank, A.G. 188–9
- freeports 216
- friction 9–10, 14
- fringing reefs 58
- fuel poverty 271
- fuelwood 270–1, 278

G

- gabions 50
- Gangotri Glacier 3
- gender dividend 141
- gentrification 168
- Geographic Information System (GIS) 309–10
- geothermal electricity 284
- glacial period 113
- glaciation 13
- glaciers 3
- global cities 218
- global civic society 197
- Global Hunger Index (2023) 262–3
- global tourism 232
- global value chains (GVCs) 226
- global warming 72, 110, 115
- globalisation 218–31, 255–7
- gorges 13
- gradient 294–5
- graphical skills 299–305
- graphs 299–305, 308
- gravitational sliding 88
- Green Revolution 254–5
- greenbelts 167, 213
- greenhouse effect 113–14, 115–16
- greenhouse gases 113–14, 117
- Greenland ice sheet 7
- gross domestic product (GDP) 179, 182, 232
- gross national income (GNI) 179–80, 184–5, 212
- gross national product (GNP) 179
- groundwater 7–9
- groundwater flow 3
- Group of 20 (G20) 222
- groynes 36, 50

H

- hard engineering 22, 49
- hazard-resistant design 23
- headlands 38, 39
- heatwaves 115–16
- Heavily Indebted Poor Countries (HIPC) Initiative 200
- High Force waterfall 5–6
- high-income countries (HICs)
 - development 184–5, 187
 - employment structures 211–12

- histograms 301–2
- Holocene 112
- housing 162–3, 168, 172
- Human Development Index (HDI) 182–3
- humidity 76
- Hurricane Andrew (1992) 58
- Hurricane Gilbert (1988) 55
- Hurricane Luis (1995) 55
- Hurricane Severity Index 54
- hurricanes *see* tropical cyclones
- hydraulic action 10, 13, 35
- hydraulic radius 4
- hydroelectric power 82, 274, 281
- hydrogen fuel cells 285
- hydrological cycle 7–8
- hydroponics 249

I

- ice ages 112–13
- ice core analysis 110–11
- ice sheets 111
 - Antarctica 7–8, 67, 72, 73
 - Greenland 7
- immigration rate 127
- impermeable rock 9
- Indonesia 201–5
- Industrial Revolution 210, 213, 214
- industry
 - and coastal areas 48
 - factors affecting 212–17
 - managing growth 169
 - sustainable 196
- inequality 162
- infant mortality rate 180–1
- infiltration 9
- interception 9
- interglacial period 113
- interlocking spurs 12
- internal displacement 142
- international aid 198–200
- International Federation of Robotics (IFR) 225
- International Organization for Migration (IOM) 150
- internet penetration rate 190
- irrigation 259
- isoline diagrams 297

J

- Jamaica, tourism 243–5

K

- kite graphs 302
- krill 70
- Kyoto Protocol (1997) 118

L

- labour 215
- labour migration 152, 221
- ladder of development 184
- lahars 93, 96
- lakes 15
- land tenure 260
- landfills 196
- landforms
 - coastal 35, 39
 - lowland 14–17
 - upland 12–14
- lava flows 96
- leisure 169
- levées 22
- life expectancy 180
- limestone 10

- line graphs 299
- literacy 180
- longshore drift 36–7, 44
- low-income countries (LICs)
 - development 184–6
 - employment structures 210, 212
 - food production 260–1
 - population structures 126, 136
 - water quality 28
- low-pressure systems 52–3

M

- magma 88
- magma chamber 96
- malaria 29, 62
- malnutrition 262
- mangroves 58, 60–2
- maps
 - atlas maps 291–2
 - base maps 292–3
 - choropleth maps 297
 - direction 294
 - distribution maps 298
 - gradient 294–5
 - isoline diagrams 297
 - references 293–4
 - route maps 298
 - scales 291, 293, 307
 - sketch maps 297
 - sphere of influence maps 298
 - symbols 295
- marine ecosystems 70–1
- marine protected area (MPA) 62
- marram grass 45–6
- mathematical skills 306–8
- mean 307–8
- meander neck 15
- median 308
- Mercalli Scale 100
- methane 111, 114
- Mexico, migration 152–4
- microcredit 201
- microplastics 28
- middle-income countries MICs)
 - development 184–5, 186–7
 - employment structures 210–11, 212
 - population structures 126, 136
- migration 142–54, 161
 - barriers to 144–5
 - categories of 142–3
 - causes of 144
 - impacts of 147–50
 - managing 150–2
 - Mexico 152–4
 - push and pull factors 145–6
- Milankovitch, Milutin 113
- Milankovitch cycle theory 113
- mineral exploitation 72
- mineral extraction 195
- mining 72, 82, 195
- Mississippi River 2–3, 22, 25–7
- mitigation 118–19
- mixed farming 247
- mobile homes 58
- mode 308
- Moment Magnitude Scale 101
- Mongolia, transnational corporations (TNCs) 228–31
- Mount Etna 107–8
- multiculturalism 161
- multiplier effects 189

N

natural gas 274, 280
neo-colonialism 188
net migration 127
new international division of labour (NIDL) 220
New York 161, 167
Niagara Falls 13
Nigeria, food insecurity 268–9
Nike 221–2
Nile River 2
non-renewable energy 270–1, 279–81
North American Free Trade Agreement (NAFTA) 222
no-till farming 266
nuclear energy 274, 280–1
nutrient cycling 82

O

obesity 252
oil 274, 279–80
one-child policy 130–1
open economies 187
organic farming 249
overgrazing 22
overland flow 3
oxbow lakes 15

P

Pacific Ring of Fire 91
package tours 233–4
Palisades 37, 44
Paris Agreement (2015) 118
pastoral farming 247, 253–4
Patagonian toothfish 70–1
penguins 70–2
per capita calorie supply 251–2
percentages 308
percolation 9
permeable rock 9
phytoplankton 70–1
pictograms 299
pie charts 300–1
pilot surveys 314
pitcher plants 78
plantations 80
Pleistocene 112
pollution
 Antarctica 73
 coastal areas 49
 rivers 28–32
 vehicles 195
population pyramids 134–5
populations
 decline 127
 demographic change 125
 demographic transition model 131–3
 dependency ratio 136
 diaspora 149
 distribution 125–6
 growth 124–9
 net migration 127
 structures 134–41
 see also migration
post-industrial societies 211
potholes, river bed 12–13
poverty 188, 271
precipitation 7–9
 Antarctica 68
 and climate change 117
 see also rainfall
primary sector 208, 212
product chain 209

protectionism 189
pyroclastic flows 93, 97

Q

quality of life 178, 216
quarrying 195
Quaternary Period 112–13
quaternary sector 209
questionnaires 314–16

R

radial graphs 303–4
rainfall 7, 19–20, 52, 76–7, 81–2
 see also precipitation
rainforests *see* tropical rainforests
range 308
rapids 5, 13, 14
ratios 308
raw materials 213
ray diagrams 303
recording tables 318
reforestation 22
refugees 142–4
renewable energy 270–1, 274, 281–5
revetments 50
Rhône River 17
Richter Scale 99
risk assessment 311–12
river cliff 14
river terrace 16
rivers
 Bradshaw model 5
 braiding 16
 channel roughness and shape 4–5
 characteristics 2–6
 deltas 3, 16–17
 deposition 6, 11, 15, 16
 discharge 4–5, 10
 drainage basins 2–3, 7
 erosion 5, 10–11
 investigating 322–3
 landforms 12–17
 living near 18–21
 meanders 14
 mouth 3
 pollution 28–32
 processes 9–11
 source 3
 transportation 11
 velocity 4–5, 10
robotics 225
Rocinha 165
rock
 impermeable 9
 permeable 9
 river erosion 5, 10–11
rock armour 50
route maps 298
rural-to-urban fringe 165–7

S

Saffir-Simpson Scale 53–4
saltation 11
sampling 312–14
sand dunes 45, 321–2
sandbanks 17
sanitary landfills 196
scales 291, 293, 307
scatter graphs 304, 308
sea breeze 37–8, 44
sea couch grasses 45–6

- sea floor spreading 93
- sea ice 111–12
 - see *also* ice sheets
- sea level change 35, 57–8, 115–16
- sea temperatures 53
- sea walls 49
- seals 70–1
- secondary sector 208, 212
- sediment 16–17, 36, 38, 43–4
- seismic waves 94
- seismograph 99
- settlements, unplanned 164
- sewage 28
- sex, and population structures 134, 137
- shale oil 280
- Shanghai, urbanisation 170–4
- shelterbelts 265
- shield volcanoes 95
- shipping 256
- Sitges 235
- sketch maps 297
- sloths 78
- smart meters 285–6
- Smoke Control Areas (SCAs) 195
- social businesses 201
- social investment 189–90
- social norms 132
- social sustainability 193
- soft engineering 22, 51
- soil
 - erosion 82, 264–5
 - salt-affected 266
 - saturated 9
 - tropical rainforests 78, 80–1
- solar power 282–3
- solution (erosion) 10
- Southern Ocean 70
- sphere of influence maps 298
- spit 43
- springs 9
- standard notation 308
- stemflow 9
- steric effect 112
- storm surges 57
- stormwater 30
- stratovolcanoes 95
- subsidence 58
- subsistence agriculture 210, 247–8
- suburbanisation 158
- sunspots 113
- supervolcano 101
- surface runoff 9
- suspended load 36
- sustainable development 191–6, 286–7
- Sustainable Development Goals 178, 191–2
- sustainable drainage systems (SuDS) 24
- Suzhou Creek 31–2
- Sweden, energy management 287–9

T

- tarrifs 197, 255–6
- technology
 - and development 190
 - disruptive 225
 - and globalisation 224–5
 - and industry 217
- tectonic activity
 - earthquakes 91–4, 98–106
 - hazards of 98–9
 - impacts of 98
 - measuring 99–101
 - plate boundaries 89–94

- predicting 103
 - responses to 102
 - volcanoes 91–7, 99–103, 107–8, 113
- Tees River 5–6
- temperature, Antarctica 68, 73
- tephra 96
- tertiary sector 208–9, 212
- Thomas Cook 233–4
- throughflow 3
- thunderstorms 52
- tidal power 283
- tied aid 199
- tomobolos 44
- tourism
 - Antarctica 73
 - Butler Model 235–6
 - coastal areas 47–8
 - economic impacts 236–7
 - ecotourism 241
 - environmental impacts 239–41
 - global 232–45
 - Jamaica 243–5
 - package tours 233–4
 - protected areas 241–2
 - seasonality 237
 - social and cultural impacts 237–9
 - sustainable 240–1
 - travel motivators 234
- traction 11
- trade 189, 197–8, 221, 236, 255–6
- trade blocs 221–2
- trade winds 37
- Trans-Amazonian Highway 80
- transnational corporations (TNCs) 225–31
- transpiration 9
- transport hubs 222–3
- transportation 11, 36, 47–8, 80, 161, 163–4, 166, 168–9, 172, 216, 256
- triangular graphs 304–5
- tributaries 2–3
- tropical cyclones 37, 44, 48, 51–7, 96, 120
 - management strategies 55–6, 58
 - Saffir-Simpson Scale 53–4
- tropical rainforests
 - climate 76–7
 - ecosystem 77
 - flora and fauna 78–9
 - food chain 79
 - global distribution 76
 - management strategies 83–4
 - threats to 80
- tsunamis 48, 99
- Typhoon Haiyan (2012) 56–7

U

- Ukraine, migration 144
- Ultra Low Emission Zone (ULEZ) 195
- United Nations Conference on Trade and Development (UNCTAD) 197
- United Nations Development Programme (UNDP) 178
- United Nations Framework Convention of Climate Change (UNFCCC) 118
- upland landforms 12
- urban fringe 165–7
- urban growth 157
- urban sprawl 165–7
- urbanisation 20, 47–8, 157–74, 253
 - challenges 162–5
 - environmental issues 170
 - managing growth 168–74
 - opportunities 161
 - rural-to-urban fringe 165–7
 - social issues 169–70
 - sustainable 195

V

- Venn diagrams 305
- visual images 309–10
- volcanic block 97
- Volcanic Explosivity Index 101
- volcanic gasses 97
- volcanoes 91–7, 99–103, 107–8, 113
- Volga River 3
- V-shaped valleys 12

W

- waste management 164, 173, 196
- waste water treatment 28–30
- water
 - cycle 7–8
 - fresh 7–8
 - quality 28–9
 - stores 7
 - stormwater 30
 - table 9
 - transfers 8
- waterfalls 5, 13–14
- watershed 2
- wave power 283
- waves 35, 37–40
- weather, impacts of climate change 115
- weight loss (manufacturing) 213
- weight-gain industries 216
- wellbeing 178
- wellfields 21
- wetted perimeter 4
- whales 70–1
- wind power 282
- wind speeds 52–3, 68
- World Trade Organization (WTO) 197–8, 221

Y

- Yangtze River 2
- yellow fever 29

Z

- zooplankton 70–1



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